

 Add icon Add cover

Copilot sparing session

1 Authoring & Domain Systems (Edge Layer)

These are not yours to replace. They are the production engines.

Examples

- Bentley Systems (OpenRoads, OpenPlant, ProjectWise ecosystem)
- Autodesk (Revit, Civil 3D, ACC)
- AVEVA (E3D, Engineering)

Role in UPE:

- Create models (BIM, 3D, P&IDs, etc.)
- Produce drawings, specs, engineering data
- Maintain tool-native metadata

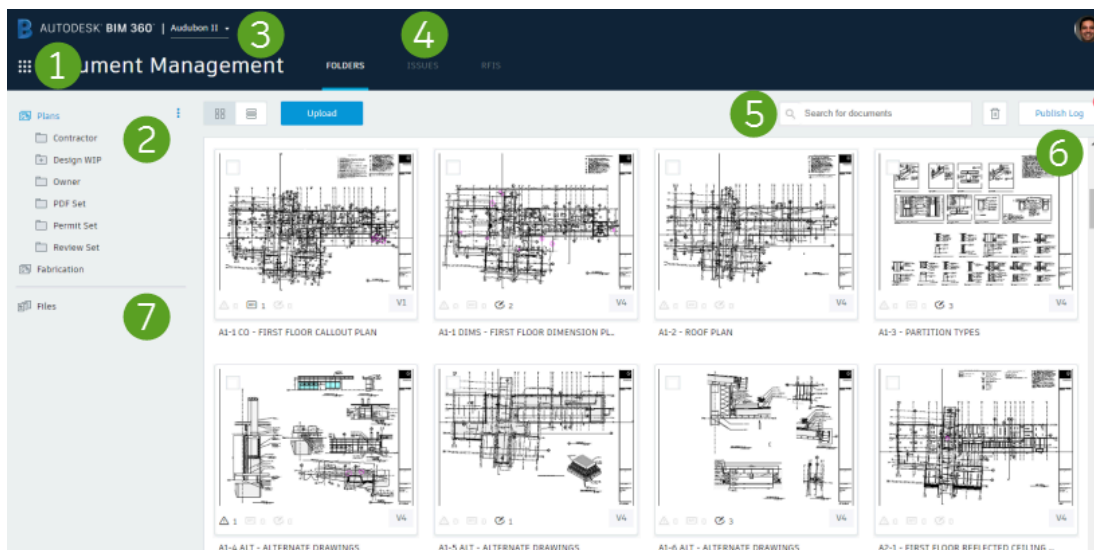
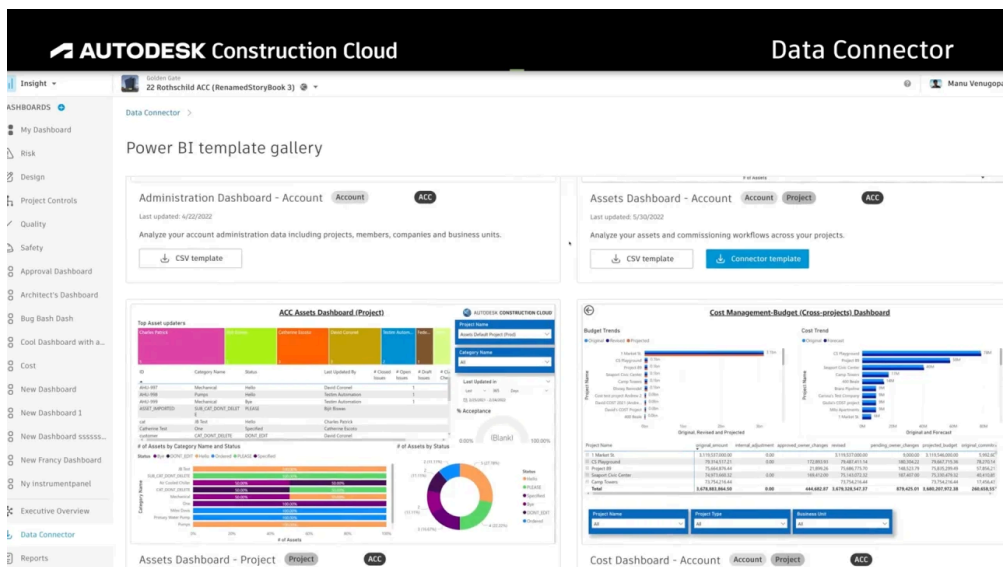
You don't disrupt this layer. You integrate it.

2 Common Data Environment (CDE) Layer

This is the **project system of record** per project.

Typical Platforms

Autodesk Construction Cloud



Bentley ProjectWise

Role:

- Version control

- Document management
- Approvals
- Design coordination
- Issue tracking

⚠ Important: CDE ≠ enterprise brain.

It's project-centric, not enterprise-centric.

3 Integration Layer (iPaaS / API Hub)

This is the connective tissue.

Examples

- MuleSoft
- Dell Boomi
- Workato

Responsibilities:

- Sync CDE ↔ ERP
- Sync CDE ↔ CRM
- Sync HR ↔ project staffing
- API normalization
- Event-driven integration

This prevents point-to-point chaos.

4 Orchestration Layer (Process Control)

This is where you coordinate multi-system workflows.

Examples

- Camunda
- Apache Airflow

Responsibilities:

- Multi-system approval flows
- Cross-platform change management
- Automated compliance steps
- Triggering downstream AI pipelines

iPaaS moves data.

Orchestration governs process state.

5 Enterprise Data & AI Layer (Strategic Core)

This is likely your most important building block.

Data Platform

Example:

- Microsoft Fabric
- Data Bricks

The screenshot displays the Microsoft Fabric pipeline editor interface. At the top, a notebook activity is highlighted with a red box, showing the 'Update Audit Pipeline Run Start' action. Below the notebook, a variable table is visible, listing various pipeline-related variables such as PipelineRunId, PipelineId, StartTimeUTC, EndTimeUTC, WorkspaceId, PipelineTriggerId, ParentPipelineRunId, PipelineCompletedSuccessfully, and Process. The 'Base parameters' section is expanded, showing a table of variables with their types and values. The 'Pipeline expression builder' panel on the right is also visible, showing the '@pipeline().RunId' expression.

Name	Type	Value
PipelineRunId	String	Value
PipelineId	String	Value
StartTimeUTC	String	Value
EndTimeUTC	String	Value
WorkspaceId	String	Value
PipelineTriggerId	String	Value
ParentPipelineRunId	String	Value
PipelineCompletedSuccessfully	Int	Value
Process	String	Value

Name	Type	Value
PipelineRunId	String	@pipeline().RunId
PipelineId	String	@pipeline().Pipeline
StartTimeUTC	String	@pipeline().TriggerTime
EndTimeUTC	String	Value
WorkspaceId	String	@pipeline().DataFactory
PipelineTriggerId	String	@pipeline().TriggerId
ParentPipelineRunId	String	@pipeline().TriggeredByPipelineRunId
PipelineCompletedSuccessfully	Int	0
Process	String	Copy blob storage tables to lakehouse files

Responsibilities:

- Enterprise data lake (OneLake-style)
- Decomposition of documents
- Metadata enrichment
- Chunking for LLM
- Vector indexing
- Knowledge graph
- Cross-project analytics

This layer:

- Pulls structured data from PLM/CDE
- Decomposes PDFs + specs
- Normalizes engineering metadata
- Creates AI-ready data assets

This is your “enterprise memory.”

6 Enterprise Systems Layer

These must connect into the same architecture:

- CRM (Sales pipeline → project initiation)
- ERP (cost control, billing)
- HR (staffing, skills, holiday)
- Finance (margin control)
- Procurement

These integrate via iPaaS and feed into the enterprise data layer.

7 Collaboration Layer (User-Facing Experience)

This is often missing.

This is where UPE becomes differentiated.

Capabilities:

- Cross-tool project dashboards
- AI copilots across projects
- Model-aware collaboration
- Multi-system issue resolution
- Real-time cross-discipline review

This layer sits ABOVE tools and CDEs.

It's not another document store.

It's a **coordination and intelligence surface**.

8 Governance (Cross-Cutting Layer)

Applies to everything:

- Identity & access management
- Data classification
- Information security
- Retention policies
- Regulatory compliance
- Auditability

Without this, enterprise rollout fails.

9 Simplified Marketecture View

You can visualize it like this:

Users / Collaboration Layer

AI & Enterprise Data Platform

Orchestration

iPaaS / API Hub

CDEs (ProjectWise / ACC)

Authoring Tools (Bentley / Autodesk / AVEVA)

Enterprise Systems (ERP / CRM / HR)

10 What Most Firms Get Wrong

1. They stop at CDE.
 2. They overinvest in integration without orchestration.
 3. They deploy AI without fixing metadata.
 4. They ignore enterprise context (cost, staffing, margin).
 5. They forget collaboration UX.
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The Strategic Insight

If Ramboll wants a real UPE:

- CDE is necessary.
- Integration is necessary.
- Orchestration is critical.
- AI-ready data layer is strategic.
- Collaboration layer is differentiating.

The real moat is not storing models.

It's **coordinating intelligence across tools, projects, and enterprise systems.**

If you want, I can next:

- Turn this into a **slide-ready executive narrative**
- Or create a **clean 1-page visual structure**
- Or pressure-test whether Ramboll should build vs. buy the top layer**

Pressure test

Good. Let's stress-test this properly.

You're effectively asking:

Should Ramboll **build** the UPE intelligence & coordination layer — or **buy/assemble** it from market components?

Let's break it down brutally.

1 What Is Actually Differentiating?

Be honest here.

NOT differentiating:

- CDE (ACC, ProjectWise)
- iPaaS plumbing
- Basic orchestration
- Data lake infrastructure
- Generic AI copilots

These are commodities or will become commodities.

If you try to build these from scratch, you waste 3–5 years and burn credibility.

Potentially differentiating:

- Cross-tool project intelligence
- AI-ready engineering decomposition logic
- Enterprise-wide project memory
- Decision-support layer across Bentley / AVEVA / Autodesk
- Margin-aware design intelligence
- Cross-discipline coordination UX

That's where value lives.

2 The Three Strategic Options

Option A — Buy Everything

Use:

- Autodesk Construction Cloud
- Bentley ProjectWise
- Microsoft Fabric
- MuleSoft

And stitch together via integration.

Pros

- Fast
- Low internal risk
- Vendor-supported

Cons

- You inherit vendor boundaries
- No unified intelligence
- No strategic control
- AI layer becomes vendor-fragmented

👉 Outcome: Ramboll becomes an advanced system integrator, not a digital leader.

Option B — Build Everything

Build:

- Custom orchestration engine
- Custom data platform
- Custom AI decomposition pipelines
- Custom collaboration UX

Pros

- Total control
- True differentiation
- Long-term moat

Cons

- Massive cost
- 3–5 year runway
- Requires elite product org
- High risk of internal politics killing it

👉 Outcome: Only works if leadership commits at CEO level.

Otherwise it dies in governance committees.

Option C — Hybrid (Most Realistic)

Buy commodity layers.

Build strategic layers.

Buy:

- CDE
- iPaaS
- Data lake infra
- Identity & security

Build:

- Engineering-specific AI decomposition pipelines
- Cross-platform knowledge graph
- Decision intelligence layer
- Unified collaboration experience
- Enterprise project memory

👉 This is likely the only viable serious path.

3 The Real Risk Nobody Mentions

Your biggest risk is not technical.

It's this:

Tool vendors will try to expand upward into your strategic layer.

- Autodesk wants to own lifecycle.
- Bentley wants to own infrastructure intelligence.

- AVEVA wants digital twin dominance.
- Microsoft wants AI layer control.

If you don't define Ramboll's "control point," someone else will.

4 What Is Ramboll's Natural Control Point?

This is the key question.

Ramboll is not:

- A software vendor
- A tool creator

Ramboll *is*:

- A multi-discipline engineering integrator
- A project delivery organization
- A margin-sensitive services business

So your strategic layer should optimize:

- Cross-discipline coordination
- Risk detection
- Margin protection
- Design decision support
- Knowledge reuse across projects

That's your moat.

Not document storage.

5 The Hard Truth

If this becomes:

"Let's create a better data platform"

It fails.

If it becomes:

"Let's reduce design rework by 12%, increase margin by 2%, and cut coordination latency by 30%"

It gets funded.

UPE must anchor to:

- Margin
- Risk
- Speed
- Talent leverage

Otherwise it's just architecture theatre.

6 What I'd Do (Blunt Version)

1. Standardize on CDE per project type.
2. Establish enterprise data lake (Fabric or equivalent).
3. Build ONE strategic AI initiative:
 - Engineering document decomposition
 - Cross-project knowledge graph
 - AI-powered coordination assistant
4. Pilot it on 3 projects.
5. Measure rework reduction.
6. Scale only if measurable.

No grand transformation narrative.

No 5-year roadmap fantasy.

Ship something painful and useful.

7 The Executive Question You Must Answer

If Ramboll does nothing:

Will Autodesk / Bentley eventually eat this layer?

If yes — you need to move.

If no — you're overengineering.